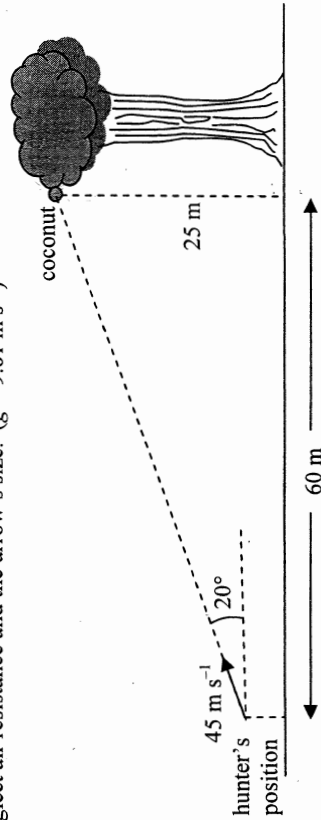


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- *(b) The hunter stands at about 60 m away from a tree as shown in Figure 5.2. He uses the bow to release the arrow in order to shoot a coconut held by a monkey (not shown in the figure) in the tree. The coconut is at a height of 25 m from the ground. The hunter aims directly at the coconut and the arrow leaves the bow at a speed of 45 m s^{-1} making an angle of 20° to the horizontal. At the moment the hunter releases the arrow, the monkey drops the coconut such that it falls vertically from rest. Neglect air resistance and the arrow's size. ($g = 9.81 \text{ m s}^{-2}$)

Figure 5.2



- (i) Find the time taken for the arrow to hit the coconut.

(2 marks)

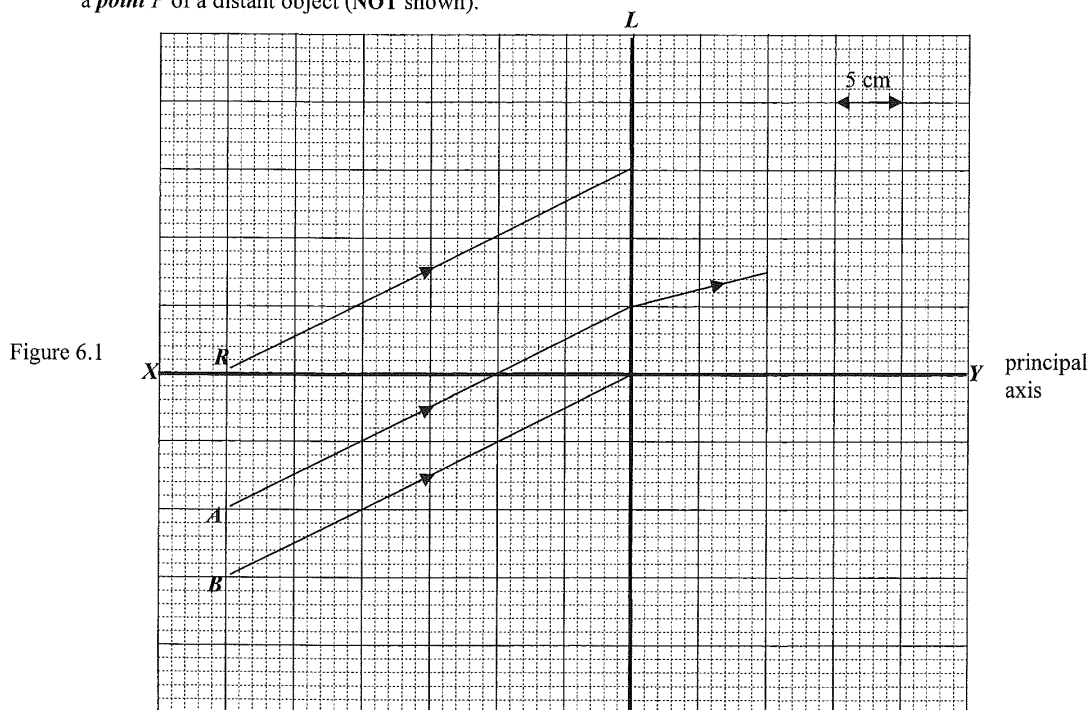
- (ii) Find the height of the coconut from the ground at the moment the arrow hits it.

(2 marks)

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6. In Figure 6.1, XY is the principal axis of a thin spherical lens L while A, B are two parallel rays coming from a point P of a distant object (NOT shown).



- (a) What kind of lens is L ? Explain. (2 marks)

- (b) (i) Locate the image of P (denoted it as point P'). (2 marks)

- (ii) Hence, determine the focal length of the lens. (1 mark)

Focal length =

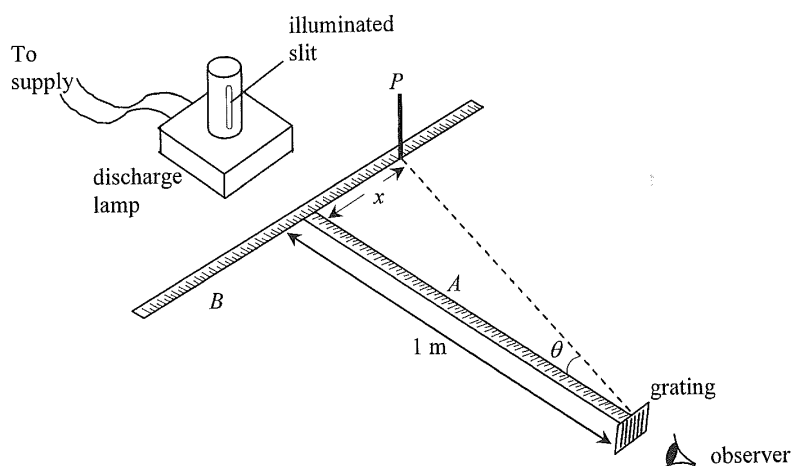
- (c) R is a ray coming from the same point P ; complete its path after passing through the lens. (1 mark)

- (d) Based on the situation shown in the ray diagram above, describe a simple experimental method to determine the focal length of lens L . (2 marks)

Answers written in the margins will not be marked.

7. Figure 7.1 shows an experimental set-up to determine the wavelength of monochromatic light emitted from the vertical narrow slit of a discharge lamp. A , B are two mutually perpendicular metre rules on the bench with rule A pointing towards the lamp. A diffraction grating with vertical lines is placed at one end of rule A . A vertically mounted pin P is moved along rule B until the pin is in line with the diffracted image of the second-order to the observer. The corresponding distance x is measured for finding the diffraction angle θ .

Figure 7.1



The grating has 300 lines per mm and x is found to be 0.38 m for the **second-order** image.

- (a) (i) Calculate the diffraction angle θ .

(1 mark)

- *(ii) Hence find the wavelength of the light from the lamp.

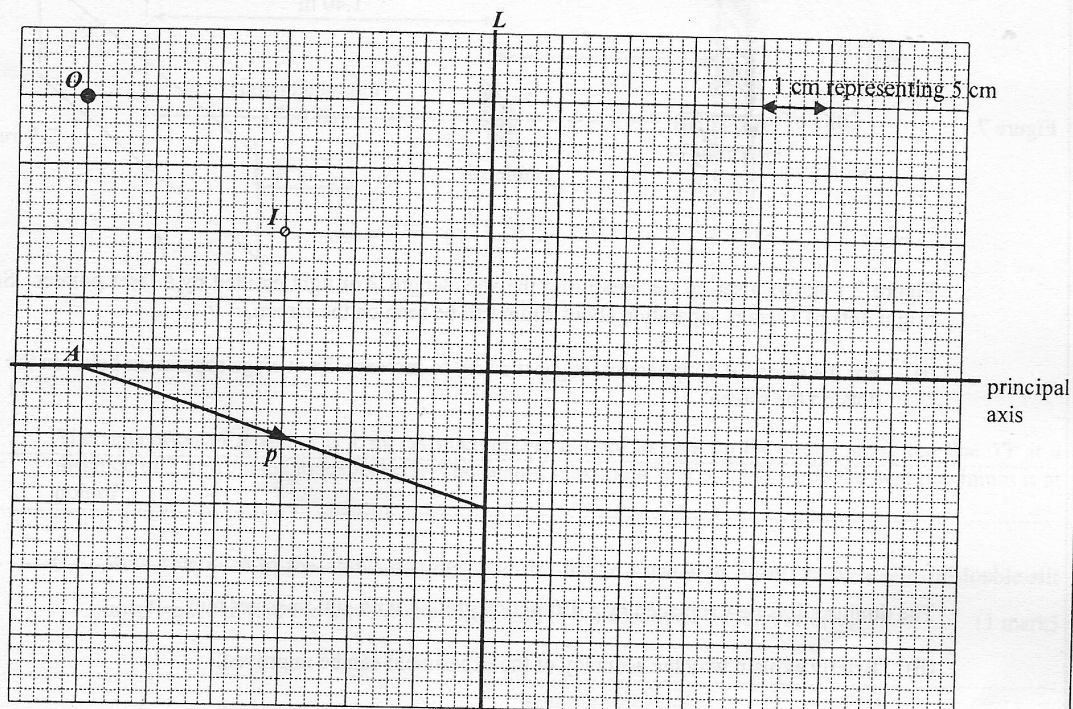
(3 marks)

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- (b) In the figure below, L represents another cylindrical lens. A vertical pin used as the object is placed at O , the image is formed at I by the lens. The horizontal scale is 1 cm to 5 cm.



- (i) What kind of lens is used? Explain.

(2 marks)

- (ii) Draw a suitable light ray to locate the principal focus F of lens L . Find its focal length.

Focal length =

(2 marks)

- (iii) Complete the path for the ray p from point A .

(1 mark)

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(iii) Only five bright dots are observed on the screen such that the separation between the 1st and 5th dots is 1.56 m. Find λ . (3 marks)

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(ii) Briefly explain why the accuracy of the experiment can be improved. (1 mark)

The double slit is now replaced by a diffraction grating with 400 lines per mm.

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(i) For the same set of apparatus, suggest a way to increase the average separation between the bright dots on the screen. (1 mark)

Figure 7.1 shows a set-up for measuring the wavelength λ of light emitted by a laser pointer. Several bright dots of average separation about 2 mm can be seen on the screen.

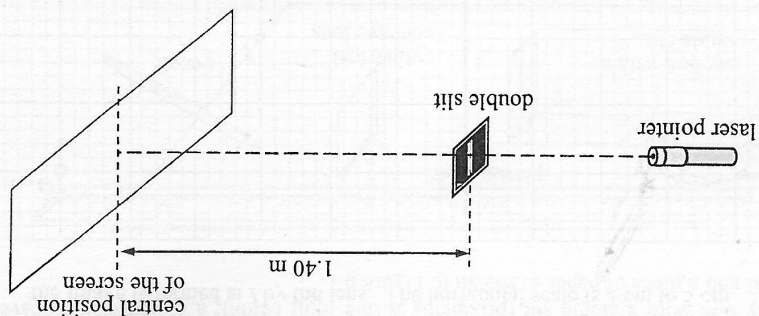


Figure 7.1

7. *(a)

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(b) The moon is orbiting around the Earth uniformly in a circular path under the influence of the Earth's gravitational attraction.

(i) Explain why the speed of the moon remains unchanged although it is acted upon by gravitational force. (2 marks)

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(ii) A student claimed that as the moon is much less massive than the Earth, it exerts negligible force on the Earth. Comment on the student's claim. (2 marks)

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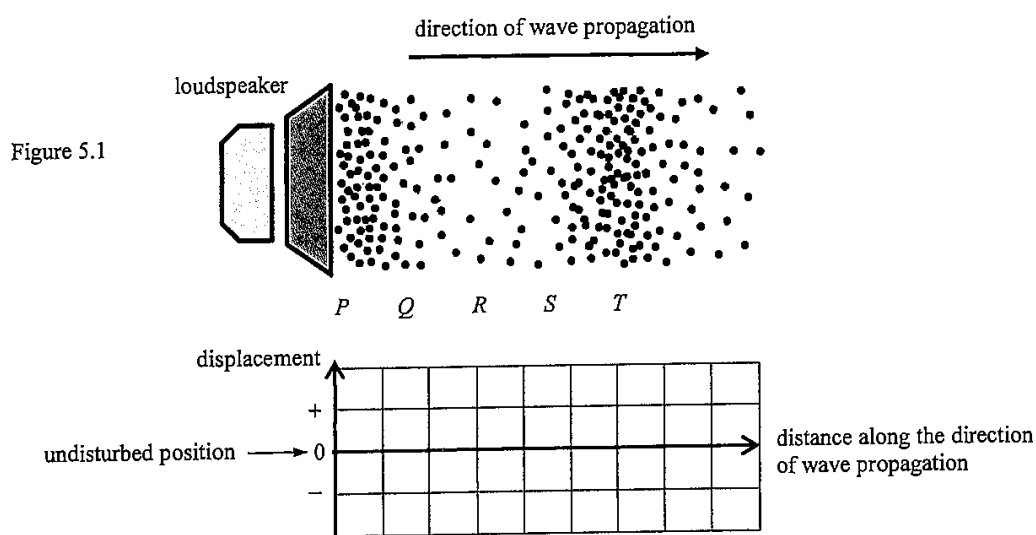
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5. Figure 5.1 represents the distribution of air particles at a certain instant in a sound wave produced by a loudspeaker.



(a) Sketch the displacement-distance graph of the air particles at that instant in the grid provided. Take the displacement to the right as positive (+). (2 marks)

(b) At the instant shown, state a region (P, Q, R, S or T) in which

(i) the air particles are moving with the highest speed; (1 mark)

(ii) the air pressure is the lowest. (1 mark)

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